RoPAWS: Robust Semi-supervised Representation Learning from Uncurated Data

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Problem of Interest

- **Semi-supervised learning (Semi-SL)**
  - Train a model using a **small labeled** data and **large unlabeled** data
  - Assumes that the unlabeled data is **i.i.d.** to the labeled data

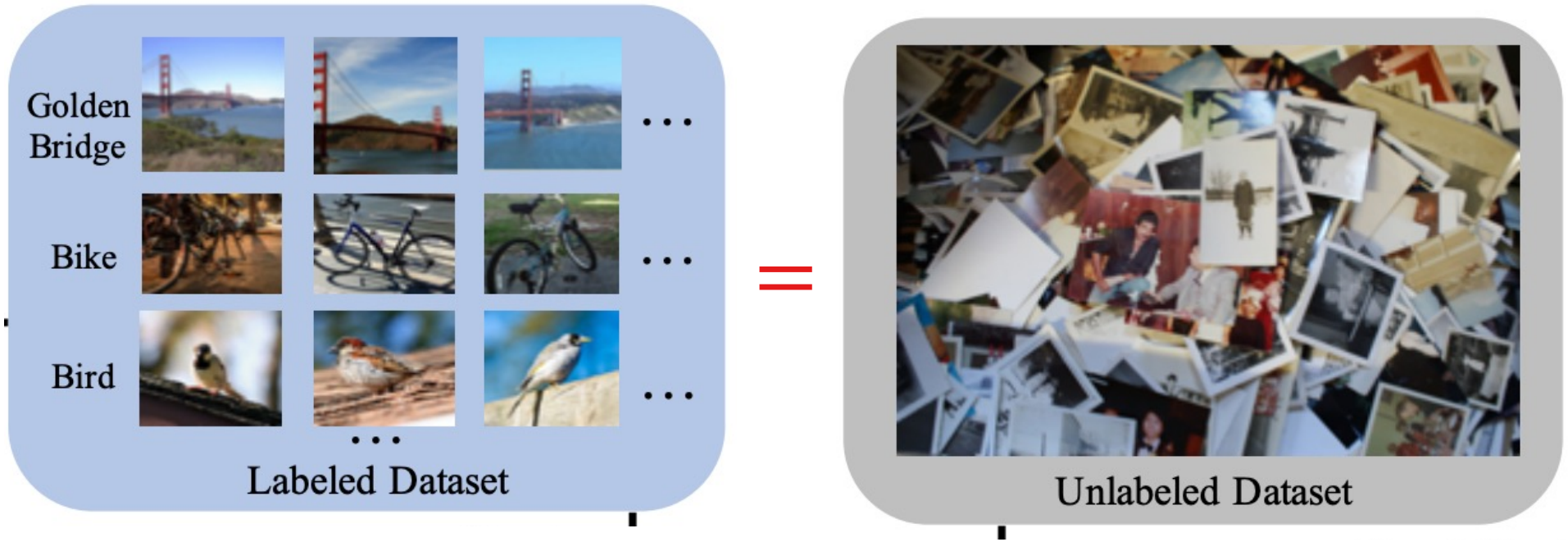


Image from Yalniz et al., "Billion-scale semi-supervised learning for image classification," arXiv'19

Problem of Interest

- **“Robust” Semi-SL**
  - Unlabeled data may come from **uncurated** sources (e.g., Google, Instagram)
  - May contain **uncertain** or **out-of-class** data, which harms Semi-SL methods

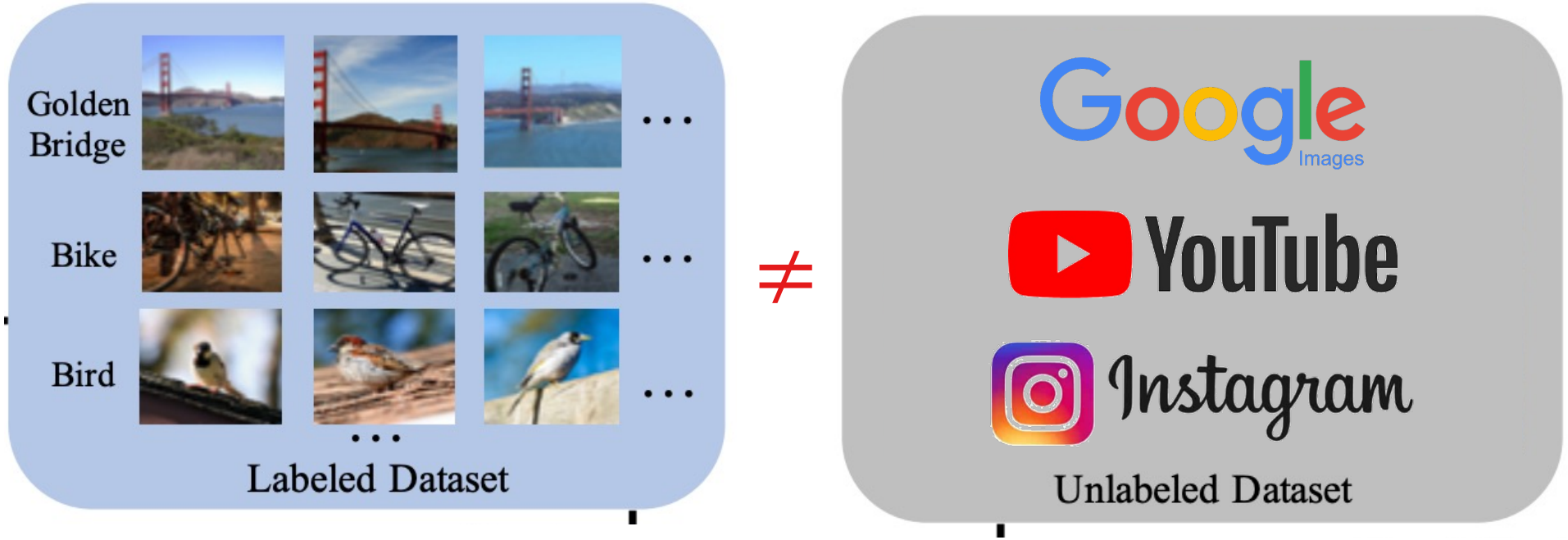


Image from Yalniz et al., “Billion-scale semi-supervised learning for image classification,” arXiv’19

Key Idea of RoPAWS

- **Issue of the SOTA method, PAWS**
  - We consider **PAWS**, the SOTA method on standard curated Semi-SL
  - PAWS often severely **fails** on the uncurated Semi-SL benchmarks

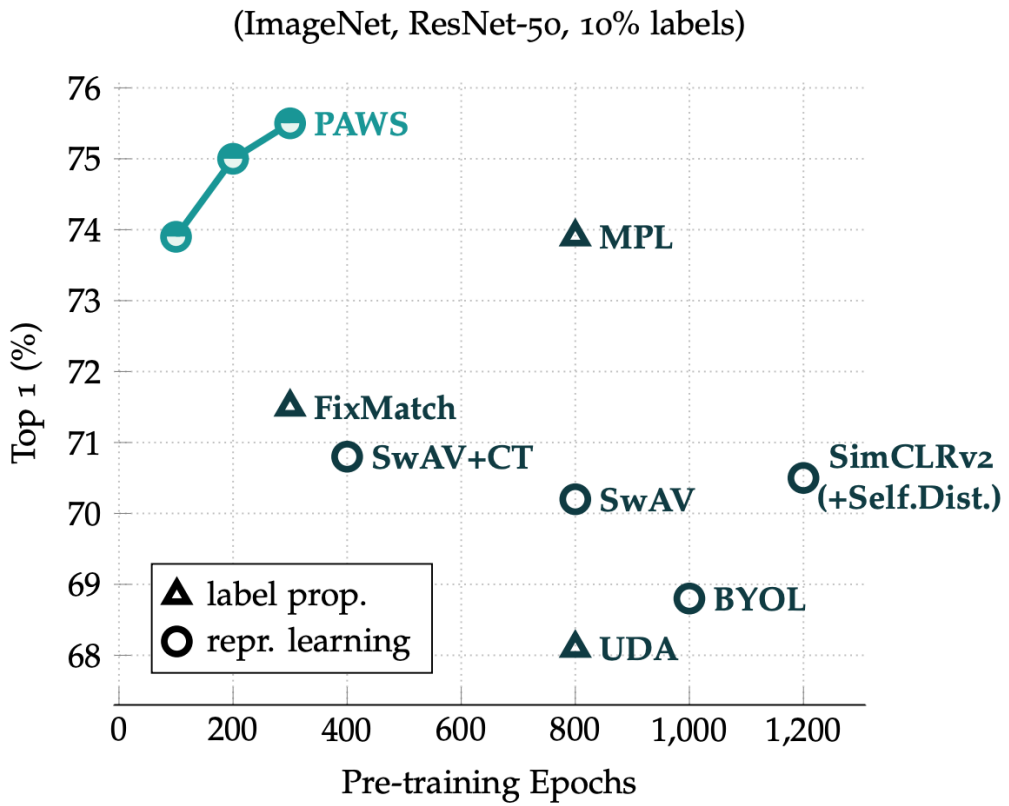
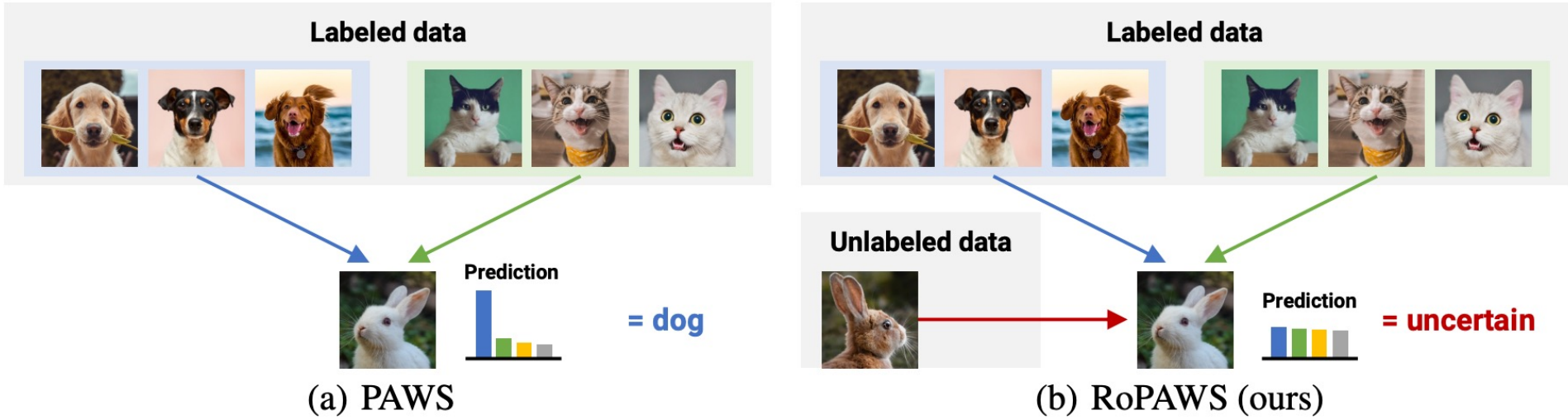


Image from Assran et al., "Semi-Supervised Learning of Visual Features by Non-Parametrically Predicting View Assignments with Support Samples," ICCV'21

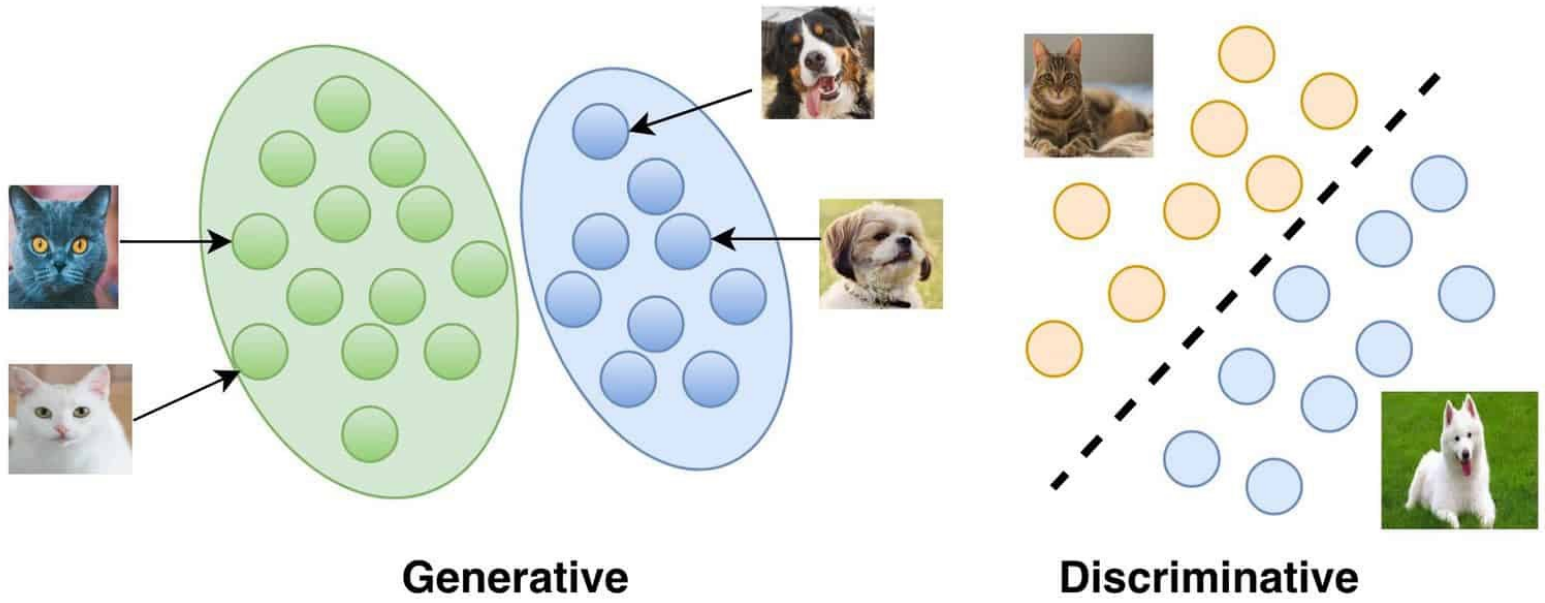
Key Idea of RoPAWS

- **How to fix PAWS?**
  - PAWS fails since it's prediction is **overconfident** to some labeled class
  - We need to **calibrate** the prediction when the unlabeled data is uncertain



Key Idea of RoPAWS

- **PAWS is secretly a generative classifier**
  - PAWS is implicitly **modeling probability** with kernel density estimation (KDE)
  - This probabilistic interpretation gives a **natural way to calibrate** the prediction



Generative

Our new probabilistic interpretation

Discriminative

Original PAWS interpretation

Image from https://learnopencv.com/generative-and-discriminative-models/

Key Idea of RoPAWS

- **RoPAWS, a novel Bayesian approach for Robust Semi-SL**
  - RoPAWS **calibrates prediction** and **regularizes out-of-class** data (see paper for details)
  - We also discuss a **novel connection** to the classic **label propagation** principle



$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Bayes' rule is all you need
for robust Semi-SL!

Main Results

- **RoPAWS achieves new SOTA on robust Semi-SL benchmarks**
  - Outperforms previous methods on standard **CIFAR + out-of-class** benchmarks

Table 8: Test accuracy (%) of various robust Semi-SL methods under various scenarios. We bold the best results. RoPAWS achieves state-of-the-art results in all considered scenarios.

In-class	CIFAR-10		CIFAR-100	
Out-of-class	SVHN	Tiny-IN	SVHN	Tiny-IN
FixMatch (Sohn et al., 2020)	68.0±0.7	70.5±1.2	41.7±1.3	46.0±1.0
PAWS (Assran et al., 2021)	47.1±0.4	60.4±5.3	36.9±0.2	46.0±0.5
UASD (Chen et al., 2020d)	66.7±1.0	74.0±0.4	39.5±0.8	44.6±0.8
RealMix (Nair et al., 2019)	58.2±5.3	69.2±2.3	44.1±1.0	47.6±1.4
DS^{3}L (Guo et al., 2020)	50.0±8.3	69.1±2.3	29.0±1.0	40.2±0.9
OpenMatch (Saito et al., 2021)	74.7±1.0	78.0±0.8	29.0±1.3	32.2±2.6
OpenCoS (Park et al., 2021)	79.0±1.1	82.5±1.3	49.2±1.2	54.0±1.8
RoPAWS (ours)	**81.2**±0.4	**83.1**±1.8	**51.5**±1.7	**59.8**±1.3

Main Results

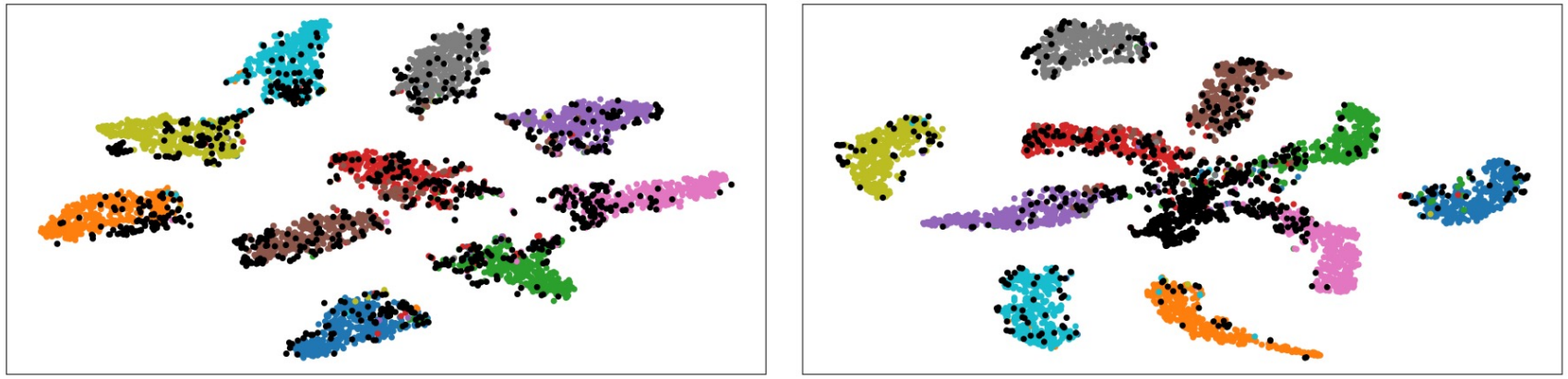
- **RoPAWS is scalable to the large-scale benchmark**
  - RoPAWS also works well on Semi-iNat, the **large-scale** robust Semi-SL benchmark

Table 1: Test accuracy (%) of Semi-SL methods trained on Semi-iNat using ResNet-50, using either a curated or uncurated (superset of curated) unlabeled data. Subscripts denote the gain over PAWS. RoPAWS improves PAWS for all scenarios, and shows that larger uncurated data outperform curated data. Also, RoPAWS outperforms previous state-of-the-art methods initialized from MoCo-v2.

	From scratch		ImageNet pre-trained	
Method	Curated	Uncurated	Curated	Uncurated
Pseudo-label	18.6	18.8	40.3	40.3
FixMatch	15.5	11.0	44.1	38.5
Self-training	20.3	19.7	42.4	41.5
FixMatch + CCSSL	-	31.2	-	-
MoCo-v2 + Fine-tuning	30.2	31.8	41.7	40.8
MoCo-v2 + Self-training	32.0	32.9	42.6	41.5
MoCo-v2 + FixMatch + CCSSL	-	41.3	-	-
PAWS	39.2	37.7	49.7	43.0
RoPAWS (ours)	39.4 (+0.2)	43.0 (+5.3)	50.3 (+0.6)	44.3 (+1.3)

Main Results

- **Analyses and ablation studies**
  - RoPAWS indeed addresses the **overconfident** issue of PAWS
  - More experiments and analyses in the paper



(a) PAWS (b) RoPAWS (ours)

Figure 4: t-SNE visualization of embeddings of (a) PAWS and (b) RoPAWS trained on CIFAR-10 (+ CIFAR-100) with 400 labels. Colored and black points are from CIFAR-10 and CIFAR-100 respectively. PAWS pushes OOD (CIFAR-100) data to the in-class (CIFAR-10) clusters. In contrast, RoPAWS pushes out-of-class data far from all the in-class clusters.

Thank you for listening! 🐾

